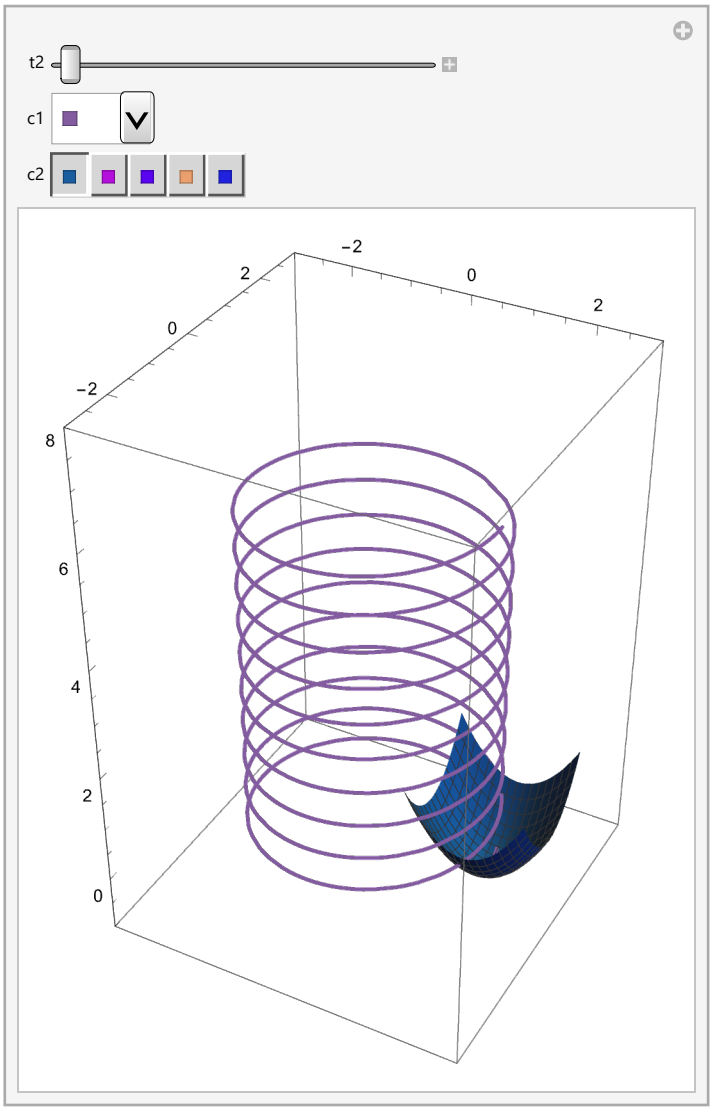
Final Exam Review

Computer Graphics, Wolfram Mathematica

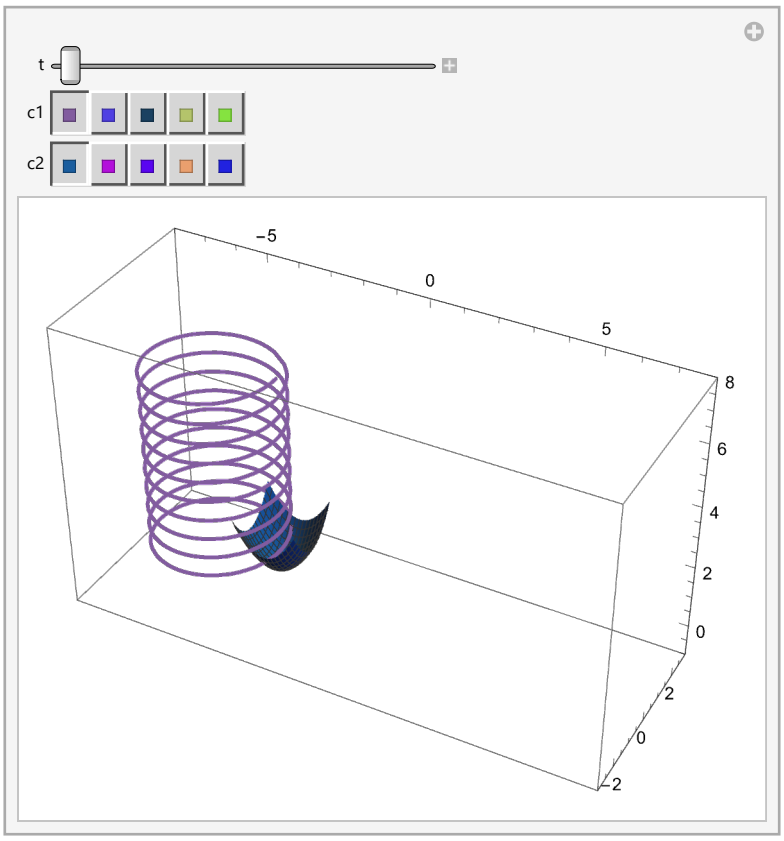
```
(* Songkran Challenge *)
```

```
Clear[obj1]
```

```
r1 = 2;
color1 = RandomColor[5]; color2 = RandomColor[5];
points1[t_] := {r1 * Cos[10 * t], r1 * Sin[10 * t], t}
obj1[t_, c_] := ParametricPlot3D[{u, v, u^2 + v^2} + points1[t], {u, -1, 1}, {v, -1, 1}, PlotStyle → c]
Manipulate[Show[ParametricPlot3D[points1[t1], {t1, 0, 2 * Pi}, PlotStyle → c1, PlotRange → {{-3, 3}, {-3, 3}, {-1, 8}},
   Axes → True], obj1[t2, c2]], {t2, 0, 2 * Pi}, {c1, color1, ControlType → PopupMenu}, {c2, color2}]
```



```
path2graphic[c_] := ParametricPlot3D[points1[t1], {t1, 0, 2 * Pi}, PlotStyle → c][[1]];
obj2graphic[t_, c_] := ParametricPlot3D[{u, v, u^2 + v^2} + points1[t], {u, -1, 1}, {v, -1, 1}, PlotStyle → c][[1]]
Manipulate[Graphics3D[
  GeometricTransformation[{path2graphic[c1], obj2graphic[(5 + t) / 10 * 2 * Pi, c2]}, TranslationTransform[{t, 0, 0}]],
  PlotRange → {{-8, 8}, {-3, 3}, {-1, 8}}, Axes → True], {t, -5, 5}, {c1, color1}, {c2, color2}]
```

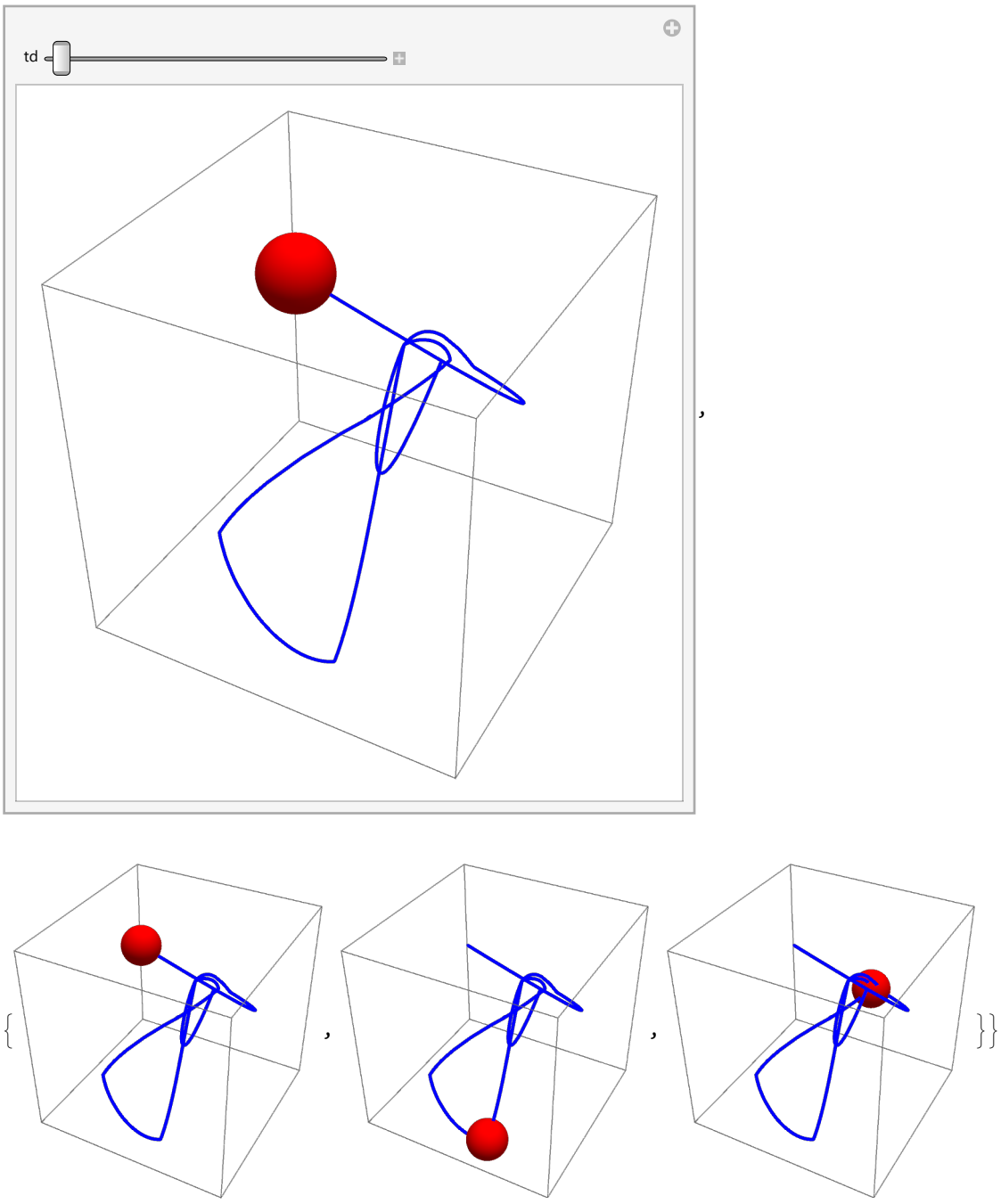



```
Interp3D[data_] := Module[{x, y, z, fx, fy, fz},
  x = Transpose[data][[1]];
  y = Transpose[data][[2]];
  z = Transpose[data][[3]];
  fx = Interpolation[x];
  fy = Interpolation[y];
  fz = Interpolation[z];
  traj = {fx[#1], fy[#1], fz[#1]} &;
  traj]
```

```
data6 = {{-4, 2, 4}, {4, 5, -1}, {2, 5, 0}, {1, 1, 3}, {2, -6, -6},
   {-4, -4, -4}, {0, 0, 0}, {0, 0, 0}, {1, 5, 0}, {1, 1, 3}, {1, -1, -1}, {2, 2, 2}};
L6 = Dimensions[data6][[1]];
```

```
MoveF6[F_, data_, ColorS_, ColorT_, R1_] := Module[{t1, SurfacePosF, plottraj, plotpoints, plotF1, plotF, Object},
  plottraj = ParametricPlot3D[Interp3D[data6][t1], {t1, 1, L6}, PlotStyle → ColorT];
  Object[t1_] := {Red, F[traj[t1], R1]};
  plotF[t1_] := Graphics3D[Object[t1]];
  plotF1[t1_] := Show[{plotF[t1], plottraj}, PlotRange → 7.5];
  plotF1]
```

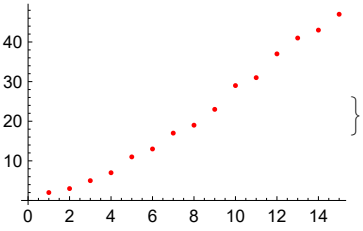
```
{Manipulate[MoveF6[Sphere, data6, Red, Blue, 1.5][td], {td, 1, L6}],
 Table[MoveF6[Sphere, data6, Red, Blue, 1.5][i], {i, 1, 12, 4}]}
```




```
(* Prime *)
```

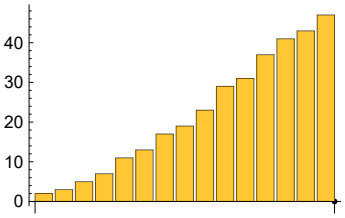
```
PrimePlot[from_, to_] := Module[{i, plot, T},
  T = Table[If[PrimeQ[i], i, Nothing], {i, from, to, 1}];
  plot = ListPlot[T, PlotStyle → {Red, PointSize[0.015]}];
  {T, plot}]
PrimePlot[1, 50]
```

{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47},



```
PrimeBar[from_, to_] := Module[{i, plot, T},
  T = Table[If[PrimeQ[i], i, Nothing], {i, from, to, 1}];
  plot = BarChart[T];
  {T, plot}]
PrimeBar[1, 50]
```

```
{{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47},
```




```
}
```

```
PrimeCount[to_] := Module[{i, T},
  T = Table[If[PrimeQ[i], i, Nothing], {i, 1, to, 1}];
  Length[T]]
```

```
PrimeList = ListPlot[Table[{i, PrimeCount[i]}, {i, 1, 100}], PlotStyle → {Red, PointSize[0.015]}];
```

```
PrimePoints = Table[{i, PrimeCount[i]}, {i, {1, 25, 50, 75, 100}}]
```

```
{{1, 0}, {25, 9}, {50, 15}, {75, 21}, {100, 25}}
```

```
PrimeFnc = Interpolation[PrimePoints, Method → "Spline"]
```

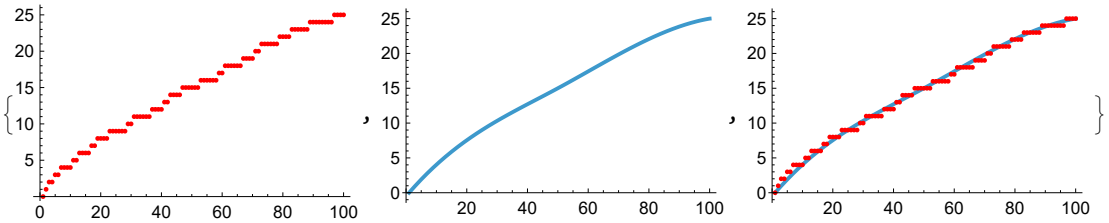
```
InterpolatingFunction[
```




```
]
```

```
PrimeGraph = Plot[PrimeFnc[t], {t, 1, 100}];
```

```
{PrimeList, PrimeGraph, Show[{PrimeGraph, PrimeList}]}
```

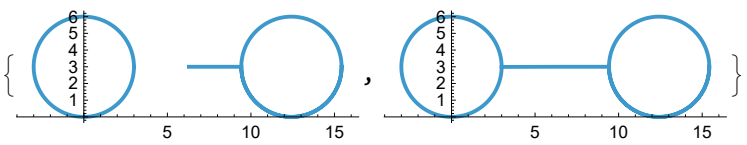



```
(* Which, Exclusion *)
```

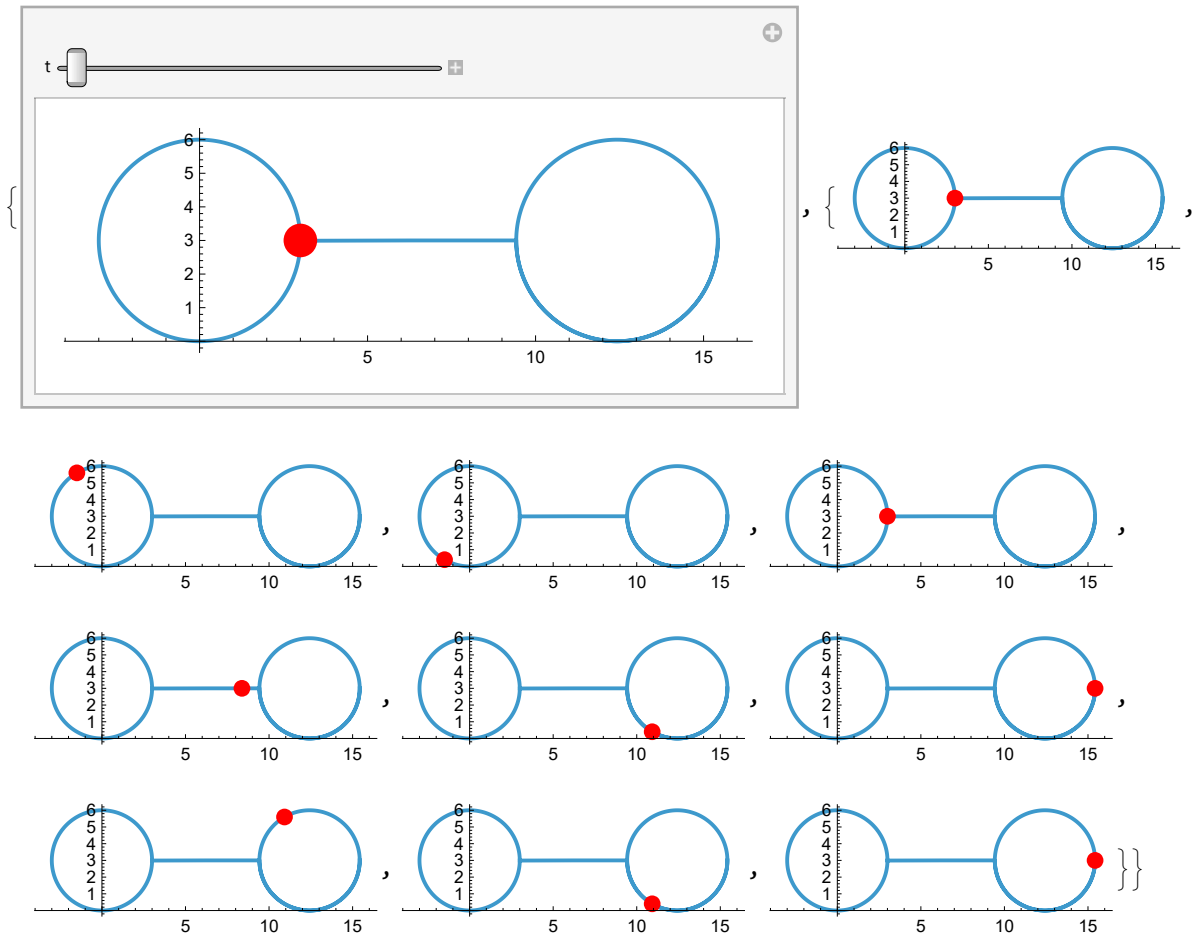
```
r4 = 3;
f1[t_] := {r4 * Cos[t], r4 * Sin[t] + r4}
f2[t_] := {t, r4}
f3[t_] := {r4 * Cos[t] + r4 + 3 * Pi, r4 * Sin[t] + r4}
```

```
path4[t_] := Which[
  t ≥ 0 && t ≤ 2 * Pi, f1[t],
  t > 2 * Pi && t < 3 * Pi, f2[t],
  t ≥ 3 * Pi, f3[t]]
```

```
{ParametricPlot[path4[t], {t, 0, 6 * Pi}],
 objpath4 = ParametricPlot[path4[t], {t, 0, 6 * Pi}, Exclusions → None]}
```



```
{Manipulate[Show[objpath4, Graphics[{Red, Disk[path4[t], 1 / 2]}, Axes → True]], {t, 0, 6 * Pi}],
 Table[Show[objpath4, Graphics[{Red, Disk[path4[t], 1 / 2]}, Axes → True]], {t, 0, 6 * Pi, Pi / 1.5}]}
```




```
(* Ladder *)
```

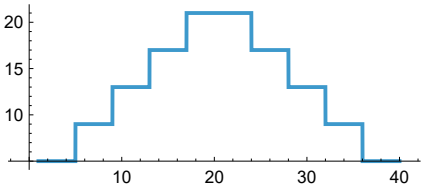
```
Ladder = {{1, 1}, {5, 5}, {9, 9}, {13, 13}, {17, 17}, {21, 21}, {24, 21}, {28, 17}, {32, 13}, {36, 9}, {40, 5}};
```

```
maxx = Max[Transpose[Ladder][[1]]];
```

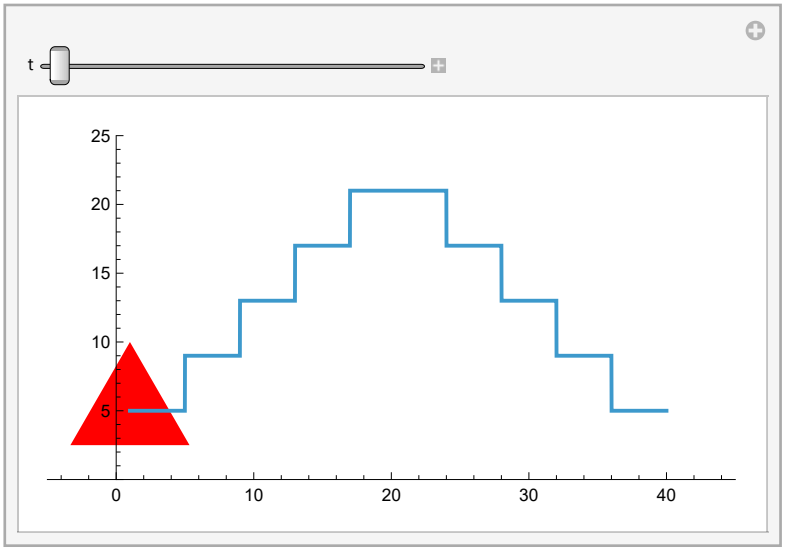
```
{fncladder = Interpolation[Ladder, InterpolationOrder → 0],
 Interpolation[Ladder, InterpolationOrder → 1]}
```

{InterpolatingFunction[Domain: {{1, 40}} Output: scalar], InterpolatingFunction[Domain: {{1, 40}} Output: scalar]}

```
path5 = ParametricPlot[{t, fncladder[t]}, {t, 1, maxx}]
```

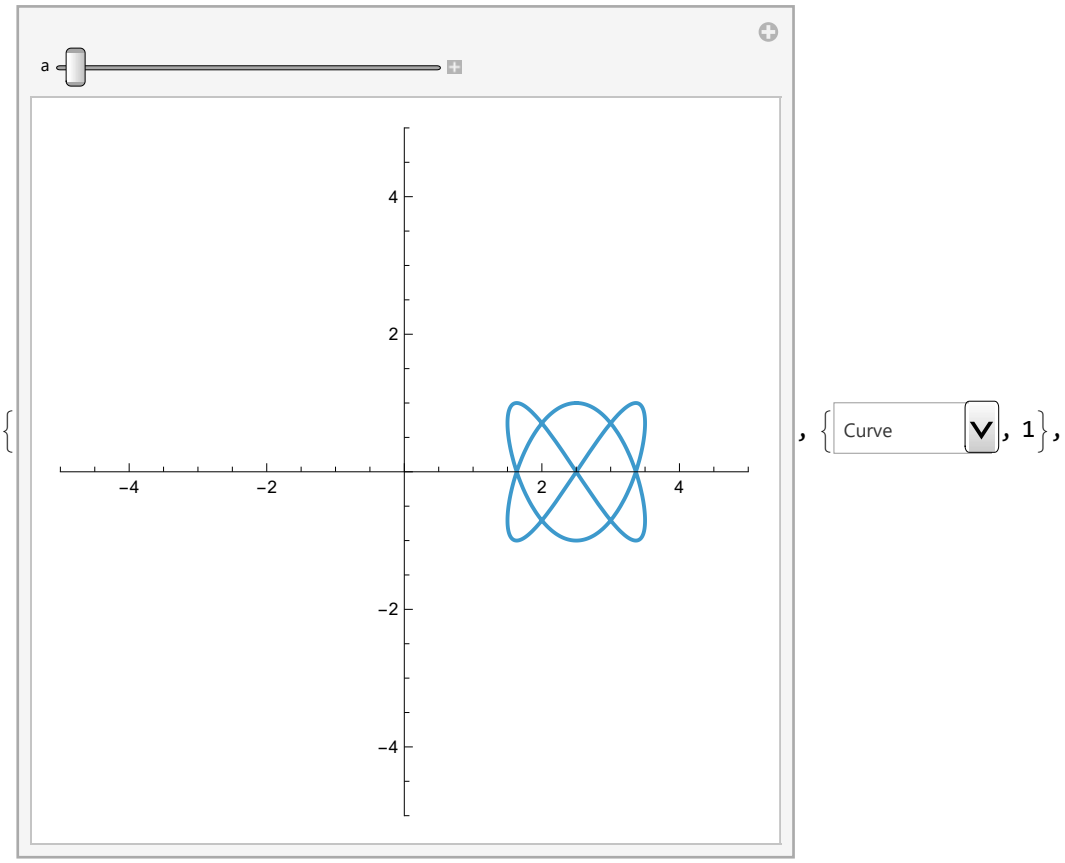


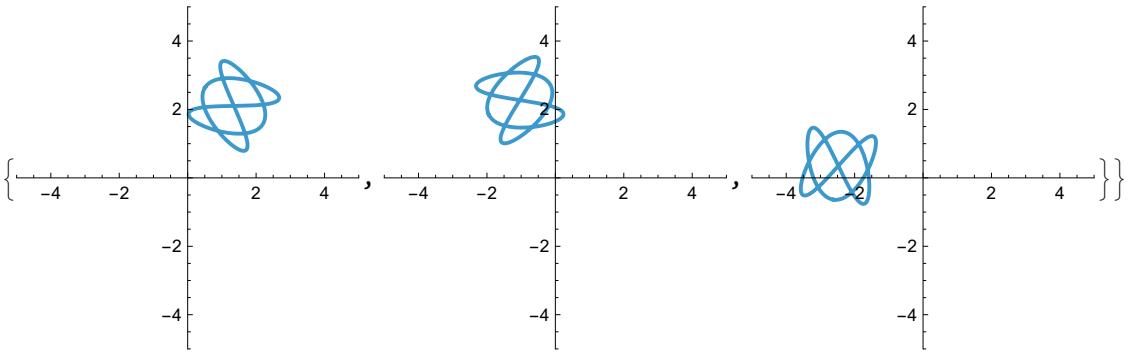
```
Manipulate[Show[{Graphics[{Red, RegularPolygon[{t, fncladder[t]}, 5, 3]},
    PlotRange → {{-5, 45}, {0, 25}}, Axes → True], path5}], {t, 1, maxx}]
```



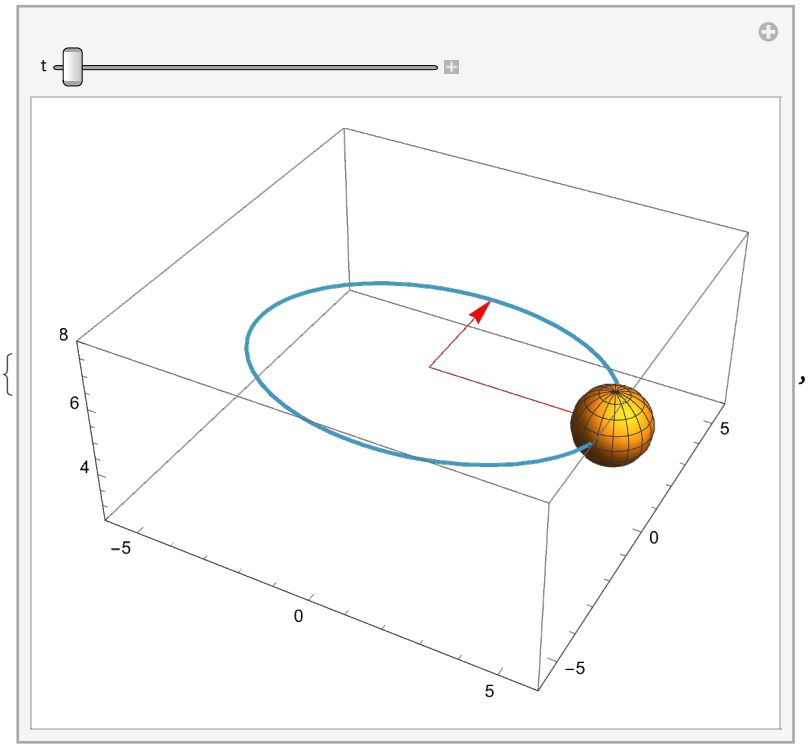

```
(* Lecture Practice *)
```

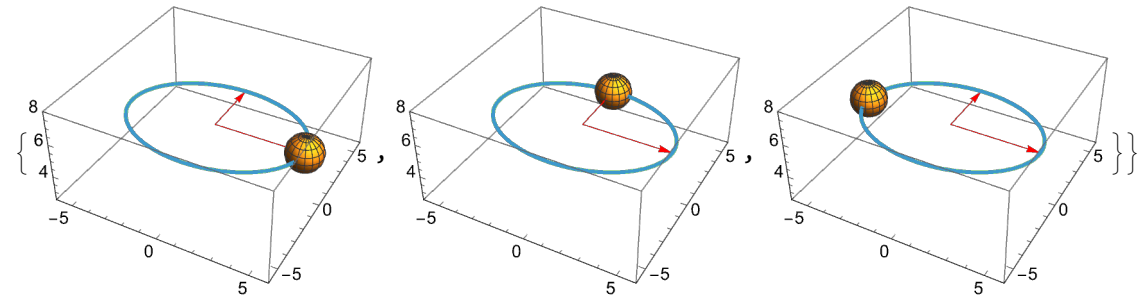
```
CurveGr1 = ParametricPlot[{Sin[2 * t], Cos[3 * t]}, {t, 0, 2 * Pi}][[1]];
Cir1 = {Blue, Thick, Circle[{0, 0}, 1.5]};
Tri1 = {RegularPolygon[{0, 0}, 1.5, 3]};
Pen1 = {RegularPolygon[{0, 0}, 1.5, 5]};
Lin1 = {Red, Thick, Line[{{-1, 0}, {1, 0}}]};
k = 1;
 Menu1 = PopupMenu[Dynamic[k], {1 → "Curve", 2 → "Circle", 3 → "Triangle", 4 → "Pentagon", 5 → "Line"}];
objlist1 = {CurveGr1, Cir1, Tri1, Pen1, Lin1};
GeoObj1[a_] :=
 GeometricTransformation[objlist1[[k]], Composition[RotationTransform[a], TranslationTransform[{2.5, 0}]]]
{Manipulate[Dynamic[Graphics[GeoObj1[a], PlotRange → 5, Axes → True]], {a, 0, 2 * Pi}], {Menu1, Dynamic[k]},
 Table[Graphics[GeoObj1[i], PlotRange → 5, Axes → True], {i, 1, 3}]}
```



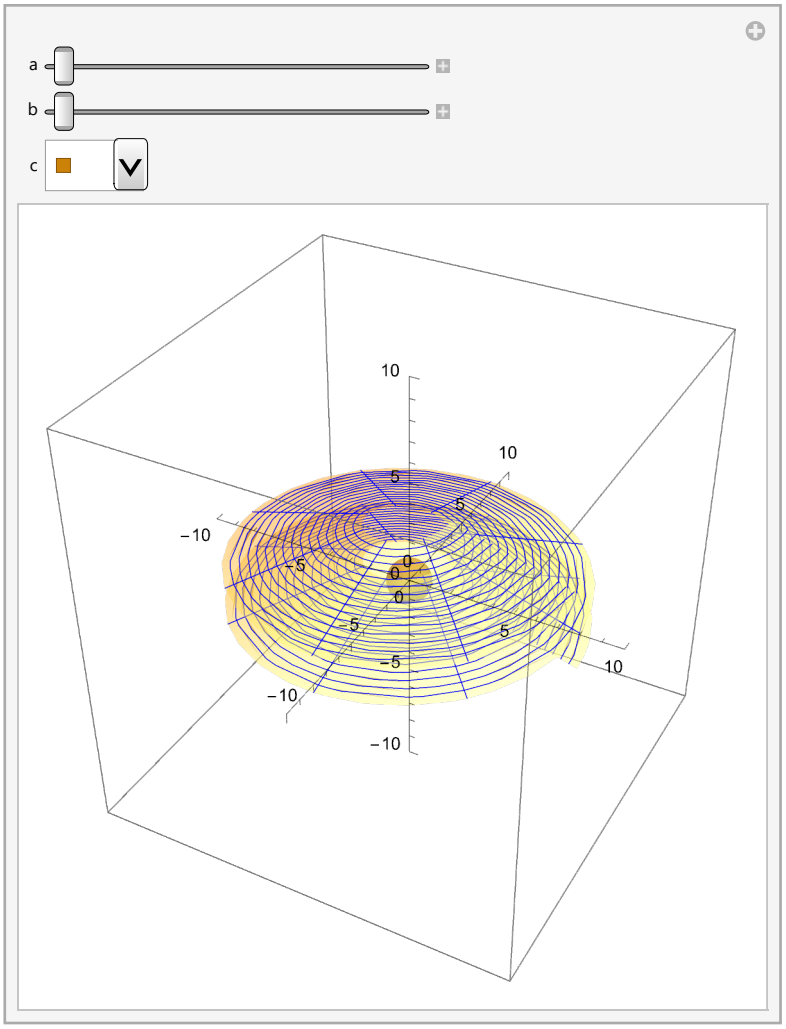


```
R = 1; R1 = 5; R2 = 3.5;
points2[t_] := {R1 * Cos[t], R2 * Sin[t], 5.5}
ecl = ParametricPlot3D[points2[t], {t, 0, 2 * Pi}, PlotRange → {{-6, 6}, {-6, 6}, {2.5, 8}}];
sph[t_] := ParametricPlot3D[
   {R * Cos[u] * Sin[v], R * Sin[u] * Sin[v], R * Cos[v]} + points2[t], {u, 0, 2 * Pi}, {v, 0, 2 * Pi}]
arr = Graphics3D[{{Red, Arrow[{{0, 0, 5.5}, {R1, 0, 5.5}}]}, {Red, Arrow[{{0, 0, 5.5}, {0, R2, 5.5}}]}}];
{Manipulate[Show[ecl, sph[t], arr], {t, 0, 2 * Pi}],
 Table[Show[ecl, sph[t], arr], {t, 0, Pi, Pi / 2}]}
```

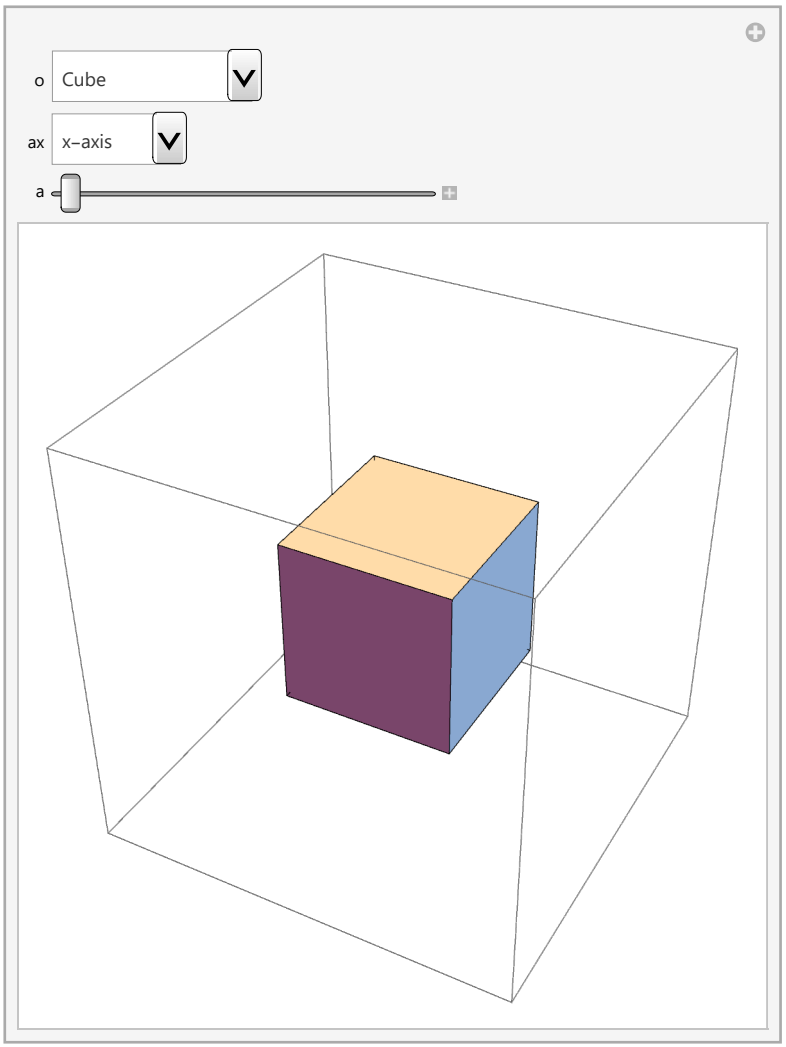




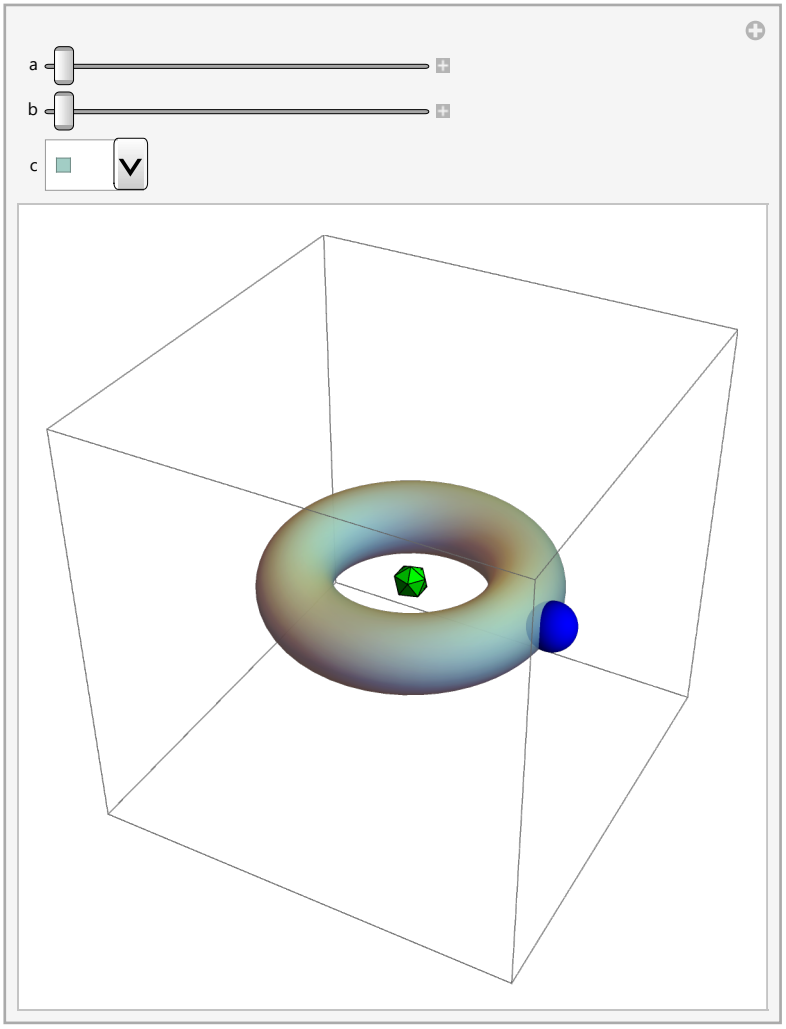
```
S[u_, v_] := {(v + 2) * Cos[u], (v + 2) * Sin[u], 0.3 * (u - v)};
objpara = ParametricPlot3D[S[u, v], {u, 0, 4 * Pi}, {v, 0, 6}, Mesh → True,
   MeshStyle → Blue, PlotPoints → 30, PlotStyle → {Orange, Specularity[5], Opacity[0.5]}];
objgraphics = objpara[[1]];
anc = {0, 0, 0};
objrotate[a_, b_] := GeometricTransformation[objgraphics, Composition[
    TranslationTransform[anc], RotationTransform[a, {0, 0, 1}], RotationTransform[b, {1, 0, 0}]]];
colors = RandomColor[12];
Manipulate[Graphics3D[{objrotate[a, b], {c, Sphere[anc, 1]}}, PlotRange → 10,
  Axes → True, AxesOrigin → {0, 0, 0}], {a, 0, 2 * Pi}, {b, 0, 2 * Pi}, {c, colors}]
```



```
obj11 = Cube[1.6];
obj2 = Dodecahedron[1];
obj3 = ParametricPlot3D[{u, v, u * v}, {u, -1, 1}, {v, -1, 1}][[1]];
objlist = {obj11 → "Cube", obj2 → "Dodecahedron", obj3 → "Surface"};
rotatelist = {{1, 0, 0} → "x-axis", {0, 1, 0} → "y-axis", {0, 0, 1} → "z-axis"};
objrotate2[o_, a_, ax_] := GeometricTransformation[o, RotationTransform[a, ax]]
Graphics3D[objrotate2[obj3, Pi / 3, {1, 0, 0}]];
Manipulate[Graphics3D[objrotate2[o, a, ax], PlotRange → 2],
 {o, objlist, ControlType → PopupMenu}, {ax, rotatelist, ControlType → PopupMenu}, {a, 0, 2 * Pi}]
```



```
tor = Torus[{0, 0, 0}, {4, 8}];
Graphics3D[{Red, tor}];
Graphics3D[tor];
sph1 = {Blue, Sphere[{0, 0, 0}, 1.3]};
Graphics3D[sph1];
cen = {Green, Icosahedron[{0, 0, 0}]};
Graphics3D[{Green, cen}];
rot1[a_] := GeometricTransformation[sph1,
  Composition[RotationTransform[a, {0, 0, 1}], TranslationTransform[{8, 0, 0}]]]
rot2[a_, b_, c_] :=
 GeometricTransformation[{{c, Opacity[0.8], tor}, rot1[a]}, RotationTransform[b, {0, 1, 0}]]
Manipulate[Graphics3D[{rot2[a, b, c], cen}, PlotRange → 12],
 {a, 0, 2 * Pi}, {b, 0, 2 * Pi}, {c, RandomColor[10]}]
```



```
knot = ParametricPlot3D[{Sin[u] + 2 * Sin[2 * u], Cos[u] - 2 * Cos[2 * u], -Sin[3 * u]},
    {u, 0, 2 * Pi}, PlotStyle → {Blue, Thickness[0.02]}];
knotgraaphics = knot[[1]];
piece[k_] := ParametricPlot3D[{Sin[u] + 2 * Sin[2 * u], Cos[u] - 2 * Cos[2 * u], -Sin[3 * u]},
   {u, k, k + 0.2}, PlotStyle → {Red, Thickness[0.03]}]
knotall[k_] := Show[knot, piece[k]]
Table[knotall[i], {i, 1, 10}];
knotgraphics[k_] := knotall[k][[1]]
Manipulate[
 Graphics3D[{{Green, PointSize[0.03], Point[{0, 0, 0}]}, GeometricTransformation[knotgraphics[t],
    RotationTransform[t, {0, 0, 1}]]}, PlotRange → 3, Axes → True], {t, 0, 2 * Pi}]
```

